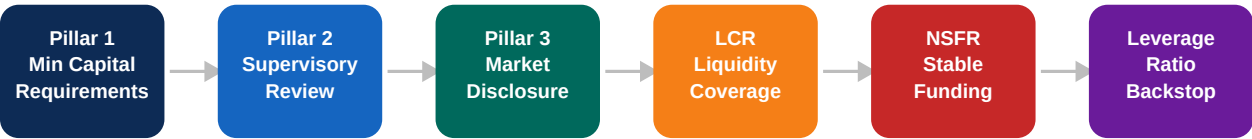


Basel III Risk Management Summary (Excerpt)

Finance Risk & Investment Intelligence System Doc
Ref: FRIIS-DOC-006 | v1.0 | 2025

This excerpt summarises key principles of the Basel III international regulatory framework as applied to the Finance Risk & Investment Intelligence System. It covers capital adequacy, risk-weighted assets, market risk, and liquidity requirements for educational simulation purposes.

Basel III Three-Pillar Framework



1. Capital Adequacy Principles

Basel III strengthened the quality and quantity of capital requirements so financial institutions can absorb losses during market stress. The capital structure consists of the following tiers:

Capital Component	Qualifying Instruments	Min Ratio (of RWA)	Quality Tier	Loss Absorption
Common Equity Tier 1 (CET1)	Ordinary shares, retained earnings, disclosed reserves	5%	Highest — going concern	First to absorb losses
Additional Tier 1 (AT1)	Perpetual bonds, non-cumulative preference shares	1.5%	High — going concern	Going-concern loss absorption
Tier 2 Capital	Subordinated debt, general loan loss provisions	2.5%	Medium — gone concern	Gone-concern absorption
Total Minimum Capital	CET1 + AT1 + Tier 2 combined	8.0%	Combined minimum	Full loss coverage
Conservation Buffer	Additional CET1 above minimum required	2.5%	CET1 quality only	Builds reserves in good times
Countercyclical Buffer (CCyB)	Jurisdiction-specific; regulator-determined	0-2.5%	CET1 quality only	Systemic risk dampener
G-SIB Surcharge	Additional buffer for systemically important banks	1-0.5%	CET1 quality only	Too-big-to-fail buffer

2. Risk-Weighted Asset (RWA) Concepts

Risk-weighted assets represent total risk exposure calculated by multiplying each asset by a risk weight reflecting the likelihood and severity of potential losses. Two primary approaches exist:

Asset Class	Standardised Risk Weight	IRB Approach	Example Assets
Cash & central bank reserves	0%	N/A	Notes, coins, CB deposits
OECD Government bonds	0%	0% (sovereign)	US Treasuries, German Bunds
Non-OECD sovereign bonds	Up to 150% (rating-based)	Model-based PD/LGD	EM government debt
Rated corporate bonds	20–100% (credit rating-based)	Own model (PD, LGD, EAD)	Investment grade corporates
Unrated corporate loans	100%	Internal model required	SME commercial loans
Residential mortgages	35–50%	Own validated model	Home loan portfolios
Commercial real estate	100–150%	Own validated model	CRE loans, property debt
Equity positions	100–250%	Own model	Listed equity, private equity
OTC Derivatives	CCF-based + CVA charge	CVA model required	Interest rate swaps, FX fws
Securitisation tranches	20–1250% (seniority-based)	SEC-IRBA or SEC-SA	CLOs, ABS, RMBS

- **Standardised Approach:** Assigns fixed risk weights from regulatory tables based on asset class and external credit rating agency ratings.
- **Foundation IRB:** Banks estimate Probability of Default (PD) using own models; LGD and EAD are set by the regulator.
- **Advanced IRB:** Banks estimate all three components (PD, LGD, EAD) using own internally validated models.
- **Basel IV Output Floor:** Internal model RWA must be at least 72.5% of standardised approach RWA under Basel IV (effective 2025).

3. Market Risk Management Overview (FRTB)

The Fundamental Review of the Trading Book (FRTB) under Basel III/IV introduces revised approaches to market risk capital. Key components include both Sensitivity-Based (SbM) and Internal Model Approaches (IMA):

Risk Component	FRTB Approach	Key Measurement	Capital Charge Basis	Desk-Level
Equity Risk	Sensitivity-based (SbM) / IMA	Delta, vega, curvature sensitivities	8% of net equity exposure	Yes — per trading desk
Interest Rate Risk	DV01 / Expected Shortfall (IMA)	Parallel shift stress scenarios	Duration × DV01 × shock factor	Yes — per desk
FX Risk	Net Open Position / SbM	VaR + Spot Rate sensitivity	8% of net long or short	Portfolio level
Commodity Risk	SbM / simplified approach	Spot and term structure risk	15% of net notional	Yes — per desk
Credit Spread Risk	Jump-to-default + spread sensitivity	Spread DV01	Spread × sensitivity factor	Yes — per desk
CVA Risk	BA-CVA or SA-CVA model	Credit valuation adjustment	Calculation of credit exposure	Portfolio level

4. Liquidity Coverage & Funding Requirements

Basel III introduced two mandatory liquidity ratios to ensure financial institutions can survive both short-term stress events and longer-term funding disruptions:

Ratio	Full Name	Minimum	Formula	Purpose	Monitoring
LCR	Liquidity Coverage Ratio	≥ 100%	HQLA / Net Cash Outflows (30-day stress)	30-day stress survival	Daily
NSFR	Net Stable Funding Ratio	≥ 100%	Available Stable Funding / Required SF	Stable funding profile	Monthly
LR	Leverage Ratio	≥ 3%	Tier 1 Capital / Total Exposure	Non-risk-based backstop cap	Quarterly
CCyB	Countercyclical Buffer	0–2.5%	Set by national macroprudential regulator	Macroprudential dampening	Adjusted
G-SIB Buffer	Systemically Important Inst.	1.0–3.5%	Based on systemic importance score	Too big-to-fail buffer	Annual review

5. Basel III Application to FRIIS Portfolio Management

While Basel III primarily governs banking institutions, its principles are widely applied to investment risk management frameworks. The following table maps Basel III concepts to the FRIIS system:

Basel III Concept	FRIIS Application	Relevant DB Field	Query Type
Capital Adequacy	Portfolio risk score thresholds and approval levels	asset_level_logs.risk_score	Hybrid: Risk + Policy
Risk-Weighted Assets	Risk-adjusted portfolio valuation model	portfolio_holdings × risk_weight	SQL + RAG
Liquidity Coverage	Minimum liquid asset requirement in portfolio	portfolio_holdings.asset_type	SQL Lookup
Stress Testing	Monthly scenario analysis across all portfolios	market_indicators + asset_risk_metrics	ML Agent
Pillar 3 Disclosure	Regulatory reporting and transparency requirements	regulatory_decisions.approval_status	Estimation Query
Countercyclical Buffer	Dynamic risk limit adjustment during market cycles	market_indicators.trend_direction	Hybrid Query
Leverage Ratio	Portfolio leverage monitoring and capital enforcement	portfolio_holdings + borrowings	SQL + Policy RAG

■ **Basel IV Note:** The Basel IV reforms (effective January 2025) further restrict internal model use, introduce the 72.5% output floor, and revise the CVA framework. The FRIIS system should accommodate both Basel III and Basel IV parameters for stress-testing and capital adequacy queries.

■ Educational excerpt only. Not a complete or legally accurate regulatory reference.